

Annexure No.:

EXPERIMENT - 08Aim :

Demonstrate the function of LCD display interfacing with Arduino.

Apparatus :

Arduino UNO, 16X2 LCD display, Breadboard, wires, potentiometer, Resistor, USB cable.

Theory :

- GND pin : need to be connected to GND (0V).
- VCC pin : the power supply for the LCD, needs to be connected to VCC (5V).

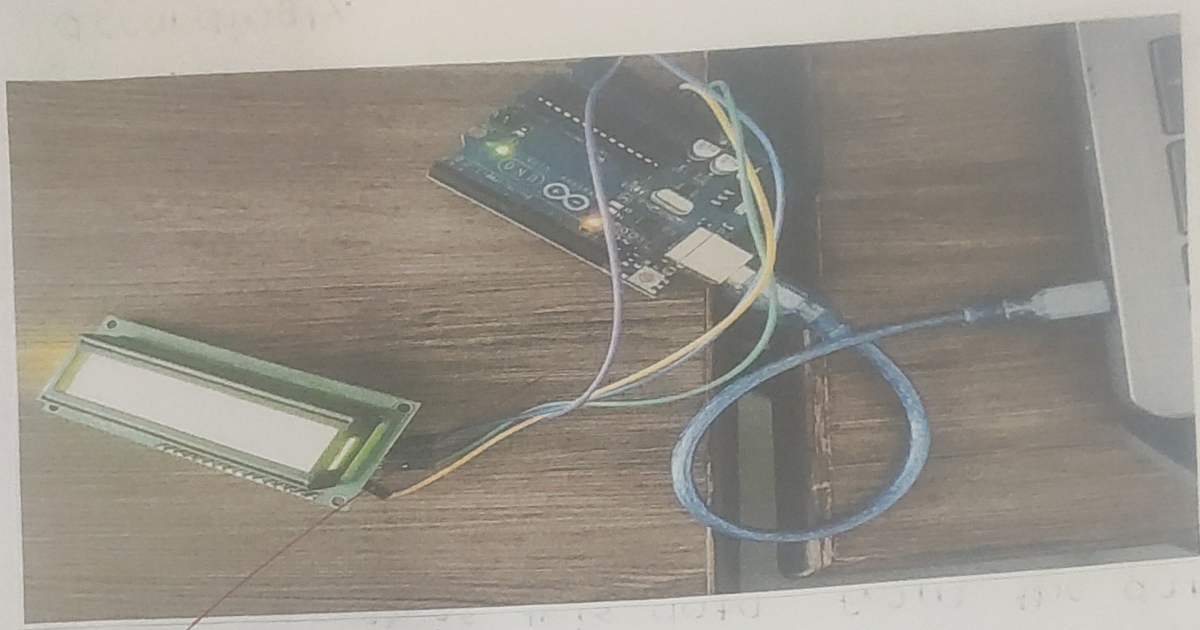
→ sending data to the LCD.

- Arduino sets the RS pin to HIGH.
- Arduino writes data to the D4-D7 pins.
- the LCD receives this data from the data bus.
- since RS is HIGH, the LCD stores the data in the data register, then displays it on the screen.

→ sending commands to the LCD (for control)

- Arduino set RS to LOW
- Arduino writes the command to the D4-D7 pins.
- the LCD receives this data from the data bus.
- since RS is LOW, the LCD stores the data in the command register and executes the command accordingly.

performance Result :



- Arduino Uno is connected to the LCD display.
- Arduino Uno is connected to the computer.
- Arduino Uno is connected to the power source.

- Need to be connected to the power source.
- Need to be connected to the computer.
- Need to be connected to the LCD display.

Conclusion :

The Arduino Uno is a microcontroller board based on the ATmega328P. It has 14 digital input/output pins (of which 6 can be used as analog inputs), 6 analog inputs, a USB Type-B connector with a built-in ATmega16U2 USB-to-UART bridge, a DC power jack, a reset button, and a 5V regulator.

Advantages :

It is easy to use, has a large community, and is very flexible. It can be used for a wide range of applications, from simple hobby projects to complex industrial systems.

Limitations :

EXPERIMENT 02

Annexure No.:

CODE :

```
#include <LCD_I2C.h>
LCD_I2C lcd (0x27, 16, 2);
void setup () {
    lcd.begin ();
    lcd.backlight ();
}
void loop () {
    lcd.print("PARUL UNIVERSITY");
    lcd.setCursor(5, 1);
    lcd.print("NAAC A++");
    delay(1000);
    for (int i=0; i<5; ++i) {
        lcd.backlight ();
        delay(50);
        lcd.noBacklight();
        delay(50);
    }
    lcd.backlight ();
    lcd.clear ();
    delay(500);
}
```

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conclusion :

LCD interfacing with Arduino UNO allows easy display of text and data, making projects more interactive and user-friendly.